UME

Organization / Business Operating System

Platform Requirements Document

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Requirements Count: 263 (198 SHALL · 52 SHOULD · 13 MAY)

Executive Summary

This Platform Requirements Document (PRD) defines what the UME Organization Operating System must do and how well it must do it. It is the authoritative reference for engineering, QA, product management, and stakeholders evaluating or building on the UME platform.

UME is a Rust-based operating system for organizations — providing a unified kernel that manages 42 domain modules covering every aspect of running an organization: strategy, finance, legal entities, HR, operations, sales, marketing, risk, compliance, IT, security, supply chain, product, and more.

Requirements Summary

Total: 263 requirements across 8 domains
SHALL (mandatory): 198 requirements — non-compliance is a defect
SHOULD (recommended): 52 requirements — deviation requires justification
MAY (optional): 13 requirements — permitted extensibility capabilities

Section 4 — Kernel:	40 requirements	covering lifecycle, RBAC, events, storage, audit
Section 5 — Modules:	85 requirements	covering all 42 organization modules
Section 6 — Interfaces:	28 requirements	covering REST API, CLI, Java client, SDK
Section 7 — Data:	19 requirements	covering models, multi-tenancy, integrity, migration
Section 8 — Security:	25 requirements	covering auth, authorization, encryption, monitoring
Section 9 — NFRs:	34 requirements	covering performance, scale, availability, usability
Section 10 — Deployment:	17 requirements	covering packaging, config, build, release
Section 11 — Extensibility:	15 requirements	covering custom modules, webhooks, integrations

Key Performance Targets

API P95 response time:	< 500 ms	Service availability:	≥ 99.9%
API P99 response time:	< 2,000 ms	Recovery time (RTO):	< 4 hours
Dashboard KPI refresh:	< 60 seconds	Recovery point (RPO):	< 1 hour
Event bus throughput:	> 10,000/sec	Concurrent users/node:	500+
Kernel boot time:	< 30 seconds	Max org scale:	100,000 employees
Audit record write:	< 50 ms	Module health check:	< 500 ms

Design Constraints

- Rust stable toolchain only — no nightly features in production code
- Domain modules MUST NOT depend on the kernel crate — only on ume_core contracts
- All external resource access MUST go through registered kernel device drivers
- Audit records are immutable — no modification after write is architecturally permitted
- RBAC enforcement occurs in the kernel — modules cannot bypass authorization
- Module fault isolation is non-negotiable — a failed module cannot take down the kernel

1. Introduction

1.1 Purpose

This document defines the platform requirements for UME — the Organization / Business Operating System. It specifies what the system shall, should, and may do to satisfy the needs of its stakeholders across all supported organizational types, scales, and deployment contexts.

Requirements are stated using RFC 2119 modal verbs: SHALL (mandatory), SHOULD (strongly recommended), and MAY (optional/extensibility). Each requirement is uniquely identified, prioritized, and accompanied by a rationale.

1.2 Scope

This document covers:

- The UME kernel and all kernel subsystems (devices, drivers, memory, storage, network, events, audit)
- All 40 built-in organization modules (Administration through Work Management)
- The custom module extension framework (UME custom + organization custom modules)
- System interfaces: HTTP API, CLI runtime, Java clients, and external integrations
- Non-functional requirements: performance, security, scalability, availability, observability
- Data architecture, persistence, and migration requirements
- Deployment, infrastructure, and operational requirements

1.3 Document Conventions

Convention	Meaning
SHALL	Mandatory requirement — non-compliance is a defect
SHOULD	Strong recommendation — deviation requires documented justification
MAY	Optional capability — permitted but not required
REQ-KRN-xxx	Kernel requirement
REQ-MOD-xxx	Organization module requirement
REQ-API-xxx	Interface / API requirement
REQ-DAT-xxx	Data architecture requirement
REQ-SEC-xxx	Security requirement
REQ-NFR-xxx	Non-functional requirement
REQ-DEP-xxx	Deployment and operations requirement
REQ-EXT-xxx	Extensibility / custom module requirement

1.4 Related Documents

Document ID	Title
UME-DA-001	UME Design & Architecture Document v1.0.0
UME-DA-002	UME Comprehensive Design & Architecture Document v1.0.0

UME-WUC-001	UME Users, Workflows & Use Cases v1.0.0
UME-WUC-002	UME Expanded Users, Workflows & Use Cases v1.0.0

2. Stakeholders & Needs

2.1 Primary Stakeholder Groups

Stakeholder	Role in UME	Key Needs
Entrepreneurs & Founders	Primary end users — operate org through UME	Speed, simplicity, compliance automation, cash visibility
Employees & Knowledge Workers	Daily users of portal, work, learning, HR modules	Unified experience, fast information retrieval, low overhead
Employers & Managers	Manage teams, budgets, performance through UME	Visibility, delegation controls, workflow automation
Independent Workers / Freelancers	Use UME as personal business OS	CRM, invoicing, time tracking, contract management
Executive Leadership (C-Suite)	Strategic oversight; board reporting	Consolidated KPIs, risk visibility, drill-down analytics
Legal & Compliance Officers	Manage entity compliance, GRC, contracts	Policy enforcement, filing schedules, audit trails
Finance Professionals	Run accounting, FP&A, treasury	Accurate ledger, automated reconciliation, multi-entity
IT Administrators & System Operators	Manage UME platform itself	Module health, config management, RBAC, backup
Software Developers	Build custom modules on UME platform	Clean SDK, stable contracts, testable interfaces
Government & Regulated Industries	Operate under public-sector rules	Transparency, audit completeness, procurement compliance
Non-Profits & INGOs	Manage grants, programs, donors	Fund accounting, impact measurement, donor reporting
Investment Funds	Manage portfolio, LPs, deal flow	Portfolio analytics, NAV computation, IC governance

2.2 Stakeholder Needs Summary

Universal Needs (all stakeholder groups)

- Single source of truth across all organizational data and processes
- Consistent, role-appropriate user experience across all modules
- Reliable, always-available system with predictable performance
- Audit-ready records for every operation — no gaps in traceability
- Configurable to fit their specific organizational context, size, and sector
- Extensible — custom modules for domain-specific needs not covered by built-ins
- Secure — data protected at rest, in transit, and at every module boundary

3. System Context & Constraints

3.1 System Boundary

UME is a Rust-based organization operating system deployed as a monorepo. The system boundary encompasses:

- The UME kernel and all kernel subsystems
- 40 built-in organization modules
- The ume_runtime CLI/REPL entry point
- The ume_server HTTP server
- The Java client suite (GUI, shell, watch modes)
- The custom module extension SDK and registry

Outside the system boundary (external dependencies via driver abstraction):

- Database backends: PostgreSQL, SQLite, Redis (via StorageDriver)
- Identity providers: OAuth2, LDAP, SAML (via IdentityDriver)
- Communication platforms: SMTP, SMS, Slack, Teams (via CommDriver)
- Event brokers: NATS, Kafka, RabbitMQ (via BrokerDriver)
- ERP and financial systems (via FinanceDriver)

3.2 Assumptions

ID	Assumption
ASM-01	The host operating system is Linux (Ubuntu 20.04+ or equivalent); Windows/macOS for development only
ASM-02	The Rust toolchain (stable channel) is available for build; minimum Rust 1.75.0
ASM-03	A minimum of 2 CPU cores and 2 GB RAM is available for the base kernel; production requires 4+ cores
ASM-04	All external system integrations are accessed exclusively via the kernel device driver abstraction
ASM-05	Custom modules are developed by trusted parties with access to the UME SDK
ASM-06	Network access to the ume_server HTTP port (default 8080) is controlled by the deploying organization
ASM-07	The Java client requires JDK 17+ and a functional Java runtime on the client machine
ASM-08	Bazel is available for Java client builds; Cargo for Rust builds

3.3 Constraints

ID	Constraint	Impact
CON-01	Rust stable toolchain only — no nightly features in production code	Limits language feature use; ensures broad toolchain compatibility

CON-02	Domain crates MUST NOT depend on the kernel crate — only on ume_core	Prevents circular dependencies; enforces modular isolation
CON-03	All external resource access MUST go through registered kernel device drivers	No direct database/network calls from modules; ensures replaceability
CON-04	Module IDs MUST be globally unique within a deployment	Prevents registry conflicts; enables safe hot-loading
CON-05	All mutating kernel facade operations MUST enforce RBAC	Security cannot be bypassed by modules; consistent authorization
CON-06	Audit records MUST be immutable after write	Compliance and legal admissibility of audit trail
CON-07	The system MUST remain operational with any single non-critical module in Error state	Fault isolation; kernel continues serving other modules
CON-08	No raw cryptographic key material may be accessed outside the SecurityDriver vault interface	Protects key material; enforces HSM-compatible key management

4. Kernel Requirements

The UME kernel is the mandatory foundation. All kernel requirements use SHALL unless explicitly noted. These requirements govern the core runtime, composition root, and all kernel subsystems.

4.1 Kernel Core & Lifecycle

Req ID	Priority	Requirement Statement	Rationale
REQ-KRN-001	SHALL	The kernel SHALL initialize as a singleton composition root; only one UmeKernel instance SHALL exist per process.	*Prevents resource conflicts and state duplication.*
REQ-KRN-002	SHALL	The kernel SHALL complete initialization within a configurable boot timeout; default 30 seconds; configurable per deployment.	*Enables health check integration with orchestrators.*
REQ-KRN-003	SHALL	The kernel SHALL load configuration from layered sources in order: defaults → config file → environment variables → runtime overrides.	*Supports all deployment patterns from dev to production.*
REQ-KRN-004	SHALL	The kernel SHALL validate all configuration on startup and reject invalid configuration with structured error messages before proceeding.	*Fail-fast prevents silent misconfiguration in production.*
REQ-KRN-005	SHALL	The kernel SHALL provide a graceful shutdown procedure that drains in-flight operations, stops modules in dependency-reverse order, and persists any buffered state.	*Data integrity on planned shutdown.*
REQ-KRN-006	SHALL	The kernel SHALL emit structured events at each lifecycle transition: booting, ready, running, degraded, shutting_down, stopped.	*Enables health monitoring and orchestrator integration.*
REQ-KRN-007	SHOULD	The kernel SHOULD provide a warm-restart capability that reinitializes configuration without terminating the process.	*Reduces downtime for configuration changes.*

4.2 Module Registry

Req ID	Priority	Requirement Statement	Rationale
REQ-KRN-010	SHALL	The kernel SHALL maintain a ModuleRegistry that registers, tracks, and manages the lifecycle of all installed organization modules.	*Central governance of all active modules.*
REQ-KRN-011	SHALL	The kernel SHALL enforce globally unique module IDs within a registry instance; duplicate registration SHALL be rejected with an error.	*Prevents identity conflicts between modules.*
REQ-KRN-012	SHALL	The kernel SHALL validate all declared module dependencies before allowing a module to transition to Running state; missing dependencies SHALL prevent startup.	*Prevents modules running without required dependencies.*
REQ-KRN-013	SHALL	The kernel SHALL support dynamic module registration at runtime without requiring kernel restart, subject to dependency validation.	*Enables zero-downtime module deployment.*
REQ-KRN-014	SHALL	The kernel SHALL expose module lookup by ID, by domain area, and by capability tag.	*Supports discovery patterns for inter-module integration.*

REQ-KRN-015	SHOULD	The kernel SHOULD maintain a module dependency graph and provide impact analysis for a given module's removal or upgrade.	*Enables safe upgrade planning.*

4.3 Supervision Engine

Req ID	Priority	Requirement Statement	Rationale
REQ-KRN-020	SHALL	The kernel SHALL monitor the health state of every registered module at a configurable interval; default 30 seconds.	*Detects module degradation before user impact.*
REQ-KRN-021	SHALL	The kernel SHALL apply a configurable restart policy to modules in Error state: max_restarts (default 3) with exponential backoff (default base 1000ms).	*Prevents restart storms while allowing auto-recovery.*
REQ-KRN-022	SHALL	The kernel SHALL transition a module to Stopped state and emit an alert when max_restarts is exceeded without successful recovery.	*Prevents infinite restart loops; escalates to operators.*
REQ-KRN-023	SHALL	The kernel SHALL isolate module failures — a module in Error or Stopped state SHALL NOT prevent other modules from operating.	*Fault isolation is a non-negotiable kernel guarantee.*
REQ-KRN-024	SHOULD	The kernel SHOULD support circuit-breaker semantics per module: open circuit after N failures, half-open probe after cooldown, close on success.	*Reduces blast radius of cascading failures.*

4.4 Device Bus & Drivers

Req ID	Priority	Requirement Statement	Rationale
REQ-KRN-030	SHALL	The kernel SHALL maintain a DeviceBus that registers, initializes, and manages the lifecycle of all device drivers.	*All external resource access is mediated by the kernel.*
REQ-KRN-031	SHALL	All device drivers SHALL implement the DeviceDriver trait: device_id, device_class, status, initialize, shutdown, health_check.	*Uniform interface enables monitoring and hot-swap.*
REQ-KRN-032	SHALL	The kernel SHALL perform a health check on all registered drivers at startup and at the configured monitoring interval.	*Detects driver failures before modules attempt to use them.*
REQ-KRN-033	SHALL	The kernel SHALL support hot-swap replacement of a driver without requiring module restart, subject to graceful drain of in-flight operations.	*Enables driver upgrades without downtime.*
REQ-KRN-034	SHALL	The kernel SHALL make no assumptions about driver availability; all module-to-driver calls SHALL handle UmeError::DriverUnavailable gracefully.	*Prevents cascading failures when external systems are unavailable.*
REQ-KRN-035	MAY	The kernel MAY support driver capability negotiation: drivers declare optional capabilities; modules query before using optional features.	*Enables graceful degradation when optional driver features are absent.*

4.5 RBAC Engine

Req ID	Priority	Requirement Statement	Rationale
REQ-KRN-040	SHALL	The kernel SHALL enforce role-based access control on all mutating operations exposed through kernel facade methods.	*Security cannot be bypassed at the module level.*
REQ-KRN-041	SHALL	The kernel SHALL evaluate RBAC can(subject, permission) before executing any mutating facade operation; denied requests SHALL return UmeError::AuthorizationFailed.	*Consistent, kernel-enforced authorization model.*
REQ-KRN-042	SHALL	The kernel SHALL write an audit record for every RBAC denial, including subject ID, attempted action, resource, and timestamp.	*Non-repudiation of denied access attempts.*
REQ-KRN-043	SHALL	RBAC role and permission definitions SHALL be loadable from configuration and SHALL support runtime modification without kernel restart.	*Enables dynamic permission management.*
REQ-KRN-044	SHALL	The kernel SHALL support role hierarchy: a subject with a parent role SHALL inherit all permissions of child roles.	*Reduces permission management overhead for hierarchical org structures.*
REQ-KRN-045	SHALL	The kernel SHALL support resource-level permission scoping: permissions may be scoped to specific resource IDs or resource classes.	*Enables fine-grained access control.*
REQ-KRN-046	SHOULD	The kernel SHOULD support permission caching with configurable TTL to reduce repeated evaluation overhead on hot paths.	*Performance optimization for high-frequency operations.*

4.6 Event Bus

Req ID	Priority	Requirement Statement	Rationale
REQ-KRN-050	SHALL	The kernel SHALL provide an event bus supporting publish/subscribe semantics for all domain events.	*Decoupled communication between modules is a core architecture principle.*
REQ-KRN-051	SHALL	Event topics SHALL follow the naming convention: {domain}.{resource}.{verb}; the kernel SHALL validate topic format on publish.	*Prevents topic naming collisions and aids discoverability.*
REQ-KRN-052	SHALL	The event bus SHALL support glob-pattern topic subscriptions (e.g., finance.* or *.created).	*Enables flexible subscription without enumerating all topics.*
REQ-KRN-053	SHALL	The event bus SHALL apply backpressure when the per-topic queue depth exceeds the configured maximum; publishers SHALL receive UmeError::BusCapacityExceeded.	*Prevents unbounded memory growth under load.*
REQ-KRN-054	SHALL	Failed event deliveries SHALL be routed to a dead-letter queue (DLQ) identified as dlq.{original_topic}; DLQ entries SHALL be inspectable and replayable.	*Ensures no events are silently lost.*
REQ-KRN-055	SHALL	The event bus SHALL be swappable via configuration: in-memory (default), NATS, Kafka, or RabbitMQ adapters.	*Supports both embedded and distributed deployment models.*
REQ-KRN-056	SHOULD	The event bus SHOULD support event replay from a cursor position for a given topic; replay SHALL be available for audit and recovery scenarios.	*Enables event sourcing and audit replay.*
REQ-KRN-057	SHOULD	The event bus SHOULD support at-least-once delivery with idempotency key deduplication.	*Prevents duplicate processing of redelivered events.*

4.7 Executor Pool

Req ID	Priority	Requirement Statement	Rationale
REQ-KRN-060	SHALL	The kernel SHALL provide named executor pools with configurable thread counts, queue depths, and priorities.	*Prevents one domain from monopolizing compute resources.*
REQ-KRN-061	SHALL	The kernel SHALL expose the following built-in executors: kernel.critical, kernel.events, kernel.audit, org.finance, org.hr, org.analytics, org.ops, org.background.	*Predefined executors cover the major domain workload classes.*
REQ-KRN-062	SHALL	Critical and background executors SHALL be isolated; a full queue in org.background SHALL NOT block kernel.critical operations.	*Kernel operations must always have compute headroom.*
REQ-KRN-063	SHOULD	The kernel SHOULD support dynamic executor scaling: thread count may be adjusted at runtime within a configured min/max bound.	*Enables adaptive scaling under varying load.*
REQ-KRN-064	SHOULD	Executor queue depths and thread utilization SHALL be exposed as kernel metrics for observability.	*Enables capacity planning and bottleneck detection.*

4.8 Memory Manager

Req ID	Priority	Requirement Statement	Rationale
REQ-KRN-070	SHALL	The kernel SHALL maintain named memory regions for domain data, event queues, cache layers, and audit buffers.	*Region-tagged allocation enables capacity management per domain.*
REQ-KRN-071	SHALL	The kernel SHALL enforce configurable per-region size limits; allocations exceeding region limits SHALL be rejected with UmeError::MemoryRegionExhausted.	*Prevents one module exhausting total process memory.*
REQ-KRN-072	SHALL	The kernel SHALL support per-region eviction policies: LRU, TTL-based, and threshold-based.	*Different data types require different eviction semantics.*
REQ-KRN-073	SHALL	The kernel SHALL expose memory utilization per region as observable metrics.	*Enables operators to detect memory pressure before OOM.*
REQ-KRN-074	SHOULD	The kernel SHOULD provide a memory pressure callback mechanism: modules may register handlers to reduce their footprint when a region approaches its limit.	*Enables cooperative memory management.*

4.9 Storage Manager

Req ID	Priority	Requirement Statement	Rationale
REQ-KRN-080	SHALL	The kernel SHALL provide a unified StorageManager API abstracting all persistence operations across all configured storage backends.	*Modules never access storage directly; backend is swappable.*
REQ-KRN-081	SHALL	The StorageManager SHALL support: get, put, delete, query, begin_transaction, commit, rollback, migrate, backup, and restore operations.	*Complete CRUD and lifecycle operation set.*

REQ-KRN-082	SHALL	All mutating StorageManager operations SHALL support ACID transactions on backends that provide transaction semantics (SQLite, PostgreSQL).	*Data integrity for all domain state changes.*
REQ-KRN-083	SHALL	The StorageManager SHALL support schema migration via versioned, idempotent migration scripts; migrations SHALL run in order and be recorded in a migration log.	*Safe schema evolution without manual intervention.*
REQ-KRN-084	SHALL	The StorageManager SHALL be configurable to use different backends per storage region: in-memory, SQLite, PostgreSQL, Redis, S3.	*Enables right-sizing storage tier per data class.*
REQ-KRN-085	SHOULD	The StorageManager SHOULD support read-replica routing: read operations may be directed to replica instances to reduce primary load.	*Scalability for read-heavy workloads.*

4.10 Log & Audit Manager

Req ID	Priority	Requirement Statement	Rationale
REQ-KRN-090	SHALL	The kernel SHALL provide a structured logging system supporting levels: TRACE, DEBUG, INFO, WARN, ERROR, CRITICAL.	*Enables environment-appropriate log verbosity configuration.*
REQ-KRN-091	SHALL	All log entries SHALL be structured (JSON-serializable) and SHALL include: timestamp (UTC RFC3339), level, module_id, message, and correlation_id.	*Enables log aggregation, filtering, and tracing across modules.*
REQ-KRN-092	SHALL	The kernel SHALL maintain an immutable append-only audit log; audit records SHALL NOT be modifiable after write.	*Legal and compliance admissibility of audit trail.*
REQ-KRN-093	SHALL	Every audit record SHALL include: id (UUID v4), at (UTC RFC3339), actor (subject_id), action ({domain}.{resource}.{verb}), resource_id, outcome (Success/Failure/Denied), session_id.	*Complete non-repudiation for every audited operation.*
REQ-KRN-094	SHALL	RBAC denial events SHALL always produce audit records regardless of module-level audit configuration; this behavior SHALL NOT be configurable to off.	*Security events must always be recorded.*
REQ-KRN-095	SHALL	Audit records SHALL be queryable by actor, action, resource_id, time range, and outcome.	*Enables compliance review and forensic investigation.*
REQ-KRN-096	SHALL	The audit log SHALL support configurable retention policies per compliance requirement; default retention SHALL be 7 years.	*Meets most regulatory minimum retention requirements.*
REQ-KRN-097	SHOULD	Audit records SHOULD be published to the event bus (topic: audit.record.created) for real-time SIEM integration.	*Enables live security monitoring without polling.*

5. Organization Module Requirements

Every organization module is a kernel-managed component implementing the KernelModule trait. This section defines (A) universal requirements that apply to all modules, then (B) per-module functional requirements for each of the 42 built-in organization modules.

5.1 Universal Module Contract (Applies to All 42 Modules)

Req ID	Priority	Requirement Statement	Rationale
REQ-MOD-001	SHALL	Every module SHALL implement the KernelModule trait, exposing: id(), domain(), version(), state(), start(), stop(), health_check(), describe(), on_event(), dependencies(), and configure().	*Uniform lifecycle management by the kernel supervisor.*
REQ-MOD-002	SHALL	Every module SHALL declare its dependencies (other module IDs and kernel services) in its ModuleDescriptor; the kernel SHALL validate all dependencies are satisfied before allowing the module to start.	*Prevents runtime dependency failures.*
REQ-MOD-003	SHALL	Every module SHALL enforce RBAC authorization on all mutating operations via the kernel RBAC facade; modules SHALL NOT implement their own authorization logic.	*Single, consistent authorization enforcement point.*
REQ-MOD-004	SHALL	Every module SHALL emit domain events on all state-changing operations via the kernel event bus; event topics SHALL follow the {domain}.{resource}.{verb} naming convention.	*Enables reactive integration between modules.*
REQ-MOD-005	SHALL	Every module SHALL write structured audit records for all create, update, delete, and approval operations via the kernel LogAuditManager facade.	*Non-repudiation for all domain operations.*
REQ-MOD-006	SHALL	Every module SHALL respond to health_check() within 500ms; a timeout SHALL cause the supervisor to record a health check failure.	*Reliable health monitoring.*
REQ-MOD-007	SHALL	Every module SHALL support graceful shutdown: on stop(), modules SHALL complete or safely abort in-flight operations and release all kernel resources within the configured drain timeout.	*Data integrity on planned shutdown.*
REQ-MOD-008	SHALL	Every module SHALL expose its public API surface exclusively through kernel facade contracts defined in ume_core; no module SHALL expose a public API that bypasses the kernel.	*Module isolation; prevents tight coupling.*
REQ-MOD-009	SHALL	Every module's data models SHALL be defined as ume_core schema types; modules SHALL NOT define schemas that mix concerns across domain boundaries.	*Schema clarity and cross-module data integrity.*
REQ-MOD-010	SHALL	Every module SHALL support runtime reconfiguration via configure(); configuration changes SHALL not require module restart unless marked restart_required in the config descriptor.	*Zero-downtime reconfiguration.*
REQ-MOD-011	SHOULD	Every module SHOULD expose Prometheus-compatible metrics for key domain operations (throughput, latency, error rate, queue depth).	*Operational observability for all domain areas.*
REQ-MOD-012	SHOULD	Every module SHOULD be independently testable via the ume_core mock kernel and mock driver interfaces without requiring a full kernel runtime.	*Enables fast, isolated module-level CI testing.*

5.2 Module-Specific Functional Requirements

5.2.1 Organization Administration (MOD-01)

Req ID	Priority	Requirement Statement	Rationale
REQ-MOD-ADM-001	SHALL	The Administration Module SHALL maintain a hierarchical organization unit (OrgUnit) tree with parent-child relationships, org unit types, and manager assignments.	*Org structure is the foundational reference for all other modules.*
REQ-MOD-ADM-002	SHALL	The Administration Module SHALL enforce referential integrity: deleting an OrgUnit with active child units or assigned employees SHALL be rejected.	*Prevents orphaned org records.*
REQ-MOD-ADM-003	SHALL	The Administration Module SHALL manage organization-wide policies with full version history; policy publication SHALL trigger a comms notification to affected units.	*Policy governance and traceability.*
REQ-MOD-ADM-004	SHALL	The Administration Module SHALL maintain the enterprise calendar: working days, public holidays by jurisdiction, and blackout periods.	*Single source of scheduling truth across all modules.*
REQ-MOD-ADM-005	SHOULD	The Administration Module SHOULD support multi-jurisdiction holiday calendars for organizations operating across multiple countries.	*Global organization support.*

5.2.2 Organization Analytics (MOD-02)

Req ID	Priority	Requirement Statement	Rationale
REQ-MOD-ANA-001	SHALL	The Analytics Module SHALL aggregate KPIs from all other modules via event subscription and direct facade queries; KPI definitions SHALL be configurable.	*Single analytics layer prevents fragmented reporting.*
REQ-MOD-ANA-002	SHALL	The Analytics Module SHALL refresh dashboards in near-real-time (within 60 seconds of an underlying data change) for all KPIs tagged as real-time.	*Executive and operational dashboards require current data.*
REQ-MOD-ANA-003	SHALL	The Analytics Module SHALL provide time-series data for all KPIs with configurable lookback period; minimum 3 years of trend history SHALL be retained.	*Trend analysis requires longitudinal data.*
REQ-MOD-ANA-004	SHALL	The Analytics Module SHALL support scheduled and ad-hoc report generation; reports SHALL be exportable to CSV, XLSX, and PDF formats.	*Meets diverse reporting workflow needs.*
REQ-MOD-ANA-005	SHOULD	The Analytics Module SHOULD provide anomaly detection alerts when a KPI deviates beyond a configured threshold from its rolling mean.	*Proactive alerting reduces reactive management.*
REQ-MOD-ANA-006	SHOULD	The Analytics Module SHOULD support embedding BI tool integrations (Metabase, Superset) via a standardized query API.	*Enables advanced analytics without replacing UME.*

5.2.3 Backup, Recovery & State Management (MOD-03)

Req ID	Priority	Requirement Statement	Rationale
REQ-MOD-BCK-001	SHALL	The Backup Module SHALL support scheduled full, incremental, and differential backups; schedules SHALL be configurable via cron expression.	*Meets diverse RPO requirements.*

REQ-MOD-BCK-002	SHALL	Every backup artifact SHALL include a BackupManifest with: artifact_path, checksum (SHA-256), backup_type, chain_ref, coverage scope, and created_at timestamp.	*Backup integrity verification and chain management.*
REQ-MOD-BCK-003	SHALL	The Backup Module SHALL verify checksum integrity of every backup artifact after write; a failed integrity check SHALL emit backup.integrity.failed and alert operators.	*Silent backup corruption must be detected immediately.*
REQ-MOD-BCK-004	SHALL	The Backup Module SHALL support restore to a point-in-time for any backup in the retention window; restore SHALL be executable to a sandbox environment for DR testing.	*Meets RTO requirements and enables DR validation.*
REQ-MOD-BCK-005	SHALL	Backup artifacts SHALL be encrypted at rest using AES-256; encryption keys SHALL be managed via the Security Module vault interface.	*Data protection for backup artifacts.*
REQ-MOD-BCK-006	SHOULD	The Backup Module SHOULD support off-site replication to a configurable secondary storage destination.	*Geographic redundancy for disaster recovery.*

5.2.4 Board Management (MOD-04)

Req ID	Priority	Requirement Statement	Rationale
REQ-MOD-BRD-001	SHALL	The Board Module SHALL manage board meetings with: meeting type, scheduled date, agenda items, attendees, quorum tracking, minutes, and resolutions.	*Complete board governance lifecycle in one module.*
REQ-MOD-BRD-002	SHALL	The Board Module SHALL track director records: appointment date, term end, roles (Chair, NED, ED), committee memberships, and declarations of interest.	*Statutory director register requirements.*
REQ-MOD-BRD-003	SHALL	The Board Module SHALL record resolutions with vote outcomes (for/against/abstained/absent); resolutions SHALL be linked to their parent agenda item.	*Non-repudiation for board decisions.*
REQ-MOD-BRD-004	SHALL	The Board Module SHALL support board pack assembly by aggregating content from Finance, Strategy, Risk, and other modules into a distributable pack.	*Automates the most time-consuming board prep task.*
REQ-MOD-BRD-005	SHALL	The Board Module SHALL enforce conflict of interest management: directors SHALL be able to declare interests; the system SHALL exclude conflicted directors from relevant items.	*Corporate governance obligation.*
REQ-MOD-BRD-006	SHOULD	The Board Module SHOULD track attendance and quorum compliance across meetings and emit alerts when quorum thresholds are at risk.	*Prevents invalid decisions due to missed quorum.*

5.2.5 Finance & Accounting (MOD-14)

Req ID	Priority	Requirement Statement	Rationale
REQ-MOD-FIN-001	SHALL	The Finance Module SHALL maintain a double-entry general ledger; all financial transactions SHALL be recorded as balanced journal entries (debits = credits).	*Fundamental accounting integrity requirement.*
REQ-MOD-FIN-002	SHALL	The Finance Module SHALL support multi-entity, multi-currency accounting; each entity SHALL have its own chart of accounts and period management.	*Multi-entity corporate structure support.*
REQ-MOD-FIN-003	SHALL	The Finance Module SHALL support period locking; once a period is locked, no journal entries SHALL be posted to that period except via an authorized adjustment entry.	*Prevents retroactive period manipulation.*

REQ-MOD-FIN-004	SHALL	The Finance Module SHALL generate GAAP/IFRS-compliant financial statements: Profit & Loss, Balance Sheet, Cash Flow Statement, and Trial Balance.	*Statutory financial reporting requirement.*
REQ-MOD-FIN-005	SHALL	The Finance Module SHALL support Accounts Payable and Accounts Receivable workflows including invoice matching, aging, dunning, and payment processing.	*Core AP/AR operational requirements.*
REQ-MOD-FIN-006	SHALL	The Finance Module SHALL perform 3-way matching of supplier invoices against purchase orders and goods receipts; match failures SHALL trigger a workflow exception.	*Prevents payment of unmatched invoices.*
REQ-MOD-FIN-007	SHALL	The Finance Module SHALL support budget management: budget lines per cost center and account, actual-vs-budget tracking, and configurable variance alerts.	*Operational budget governance.*
REQ-MOD-FIN-008	SHALL	The Finance Module SHALL support restricted fund accounting: funds SHALL be tagged as restricted or unrestricted; expenditure SHALL be prevented from exceeding a restricted fund balance.	*Non-profit and grant fund management requirement.*
REQ-MOD-FIN-009	SHALL	The Finance Module SHALL compute group consolidation: intercompany elimination, minority interest calculation, and consolidated group financial statements.	*Multi-entity group reporting requirement.*
REQ-MOD-FIN-010	SHOULD	The Finance Module SHOULD provide a 13-week rolling cash flow forecast derived from confirmed AR, AP, payroll, and committed expenditure.	*Proactive cash management for all org types.*
REQ-MOD-FIN-011	SHOULD	The Finance Module SHOULD support integration with banking platforms via ISO 20022 or Open Banking APIs for automated bank reconciliation.	*Reduces manual reconciliation effort.*

5.2.6 Human Resources (MOD-16)

Req ID	Priority	Requirement Statement	Rationale
REQ-MOD-HR-001	SHALL	The HR Module SHALL maintain a complete employee lifecycle record: pre-start, active, leave, terminated; all status transitions SHALL be audited.	*Complete employment record for all lifecycle stages.*
REQ-MOD-HR-002	SHALL	The HR Module SHALL manage recruitment workflows: requisition, sourcing, application, screening, interview, offer, and hire stages.	*Full talent acquisition lifecycle.*
REQ-MOD-HR-003	SHALL	The HR Module SHALL compute payroll: gross pay, statutory deductions, net pay, employer costs; payroll SHALL be jurisdiction-aware for tax and contribution rules.	*Core employer compliance requirement.*
REQ-MOD-HR-004	SHALL	The HR Module SHALL manage leave: leave types, balances, requests, approvals, carry-over rules, and calendar blocking.	*Employee entitlement management.*
REQ-MOD-HR-005	SHALL	The HR Module SHALL support performance management cycles: review creation, self-assessment, manager review, calibration, and final rating workflows.	*Structured performance management.*
REQ-MOD-HR-006	SHALL	The HR Module SHALL track compliance certifications and mandatory training completion per employee; overdue items SHALL trigger automated escalation.	*Employment law and safety compliance.*
REQ-MOD-HR-007	SHOULD	The HR Module SHOULD provide workforce analytics: headcount trends, turnover rates, diversity metrics, time-to-hire, and cost-per-hire.	*People analytics for strategic workforce management.*

REQ-MOD-HR-008	SHOULD	The HR Module SHOULD support succession planning: 9-box grid assessments, readiness ratings, and development plan tracking for key roles.	*Talent continuity planning.*

5.2.7 Legal Entity Management — Chombo (MOD-13)

Req ID	Priority	Requirement Statement	Rationale
REQ-MOD-CHM-001	SHALL	Chombo SHALL maintain a legal entity registry with: name, type, jurisdiction, registration number, entity status, parent entity, and ownership percentage.	*Single source of truth for all legal entities in the group.*
REQ-MOD-CHM-002	SHALL	Chombo SHALL execute a compliance evaluation (evaluate()) for any entity against all applicable policy packs; evaluation SHALL return ComplianceAlerts and FilingSchedule.	*Automated compliance gap identification.*
REQ-MOD-CHM-003	SHALL	Chombo SHALL ship with a minimum of 45 built-in jurisdiction policy packs covering major global markets; new packs SHALL be deployable without kernel restart.	*Broad jurisdictional coverage from day one.*
REQ-MOD-CHM-004	SHALL	Chombo SHALL maintain a statutory filing schedule per entity; upcoming filings SHALL be surfaced as alerts at the configured lead_days in advance.	*Prevents missed statutory deadlines.*
REQ-MOD-CHM-005	SHALL	Chombo SHALL model the complete group entity graph: parent/subsidiary/JV relationships with ownership percentages; the graph SHALL be traversable for consolidation purposes.	*Multi-entity group structure management.*
REQ-MOD-CHM-006	SHALL	Chombo SHALL emit chombo.filing.due, chombo.filing.overdue, and chombo.entity.status_changed events; overdue filings SHALL trigger escalation to configured recipients.	*Automated compliance alerting.*
REQ-MOD-CHM-007	SHOULD	Chombo SHOULD support jurisdiction-specific beneficial ownership (PSC/UBO) register management with change tracking and disclosure management.	*Meeting transparency register obligations globally.*

5.2.8 Governance, Risk & Compliance (MOD-15)

Req ID	Priority	Requirement Statement	Rationale
REQ-MOD-GRC-001	SHALL	The GRC Module SHALL maintain a compliance obligation register: obligations by domain, applicable regulation, jurisdiction, evidence requirements, and review frequency.	*Structured compliance management framework.*
REQ-MOD-GRC-002	SHALL	The GRC Module SHALL manage a control library: controls linked to obligations, design assessments, operating effectiveness ratings, and test results.	*Controls-based compliance assurance.*
REQ-MOD-GRC-003	SHALL	The GRC Module SHALL support audit management: engagement creation, scope, evidence collection, findings, management responses, and report publication.	*Full internal audit lifecycle.*
REQ-MOD-GRC-004	SHALL	The GRC Module SHALL automate evidence collection via integration with other modules: pulling control evidence from HR, Finance, IT, and Security automatically.	*Reduces manual evidence gathering burden.*
REQ-MOD-GRC-005	SHOULD	The GRC Module SHOULD support regulatory change management: ingesting new regulations, mapping to impacted obligations and controls, and generating remediation tasks.	*Proactive regulatory change response.*

5.2.9 Risk Management (MOD-33)

Req ID	Priority	Requirement Statement	Rationale
REQ-MOD-RSK-001	SHALL	The Risk Module SHALL maintain an enterprise risk register with: risk ID, category, description, likelihood, impact, inherent score, treatment type, residual score, owner, and executive sponsor.	*Complete risk record structure for enterprise risk management.*
REQ-MOD-RSK-002	SHALL	The Risk Module SHALL compute risk scores from likelihood × impact matrices; scoring matrices SHALL be configurable per organization.	*Organization-specific risk appetite calibration.*
REQ-MOD-RSK-003	SHALL	The Risk Module SHALL manage Key Risk Indicators (KRIs) with configurable amber and red thresholds; threshold breaches SHALL immediately emit risk.kri.amber_threshold or risk.kri.red_threshold events.	*Proactive risk monitoring via leading indicators.*
REQ-MOD-RSK-004	SHALL	The Risk Module SHALL manage treatment plans: mitigation actions, linked controls, target residual score, and milestone due dates.	*Closed-loop risk treatment management.*
REQ-MOD-RSK-005	SHALL	Risks exceeding the configured risk appetite threshold SHALL auto-escalate to the designated executive sponsor and Risk Committee; escalation SHALL be audited.	*Automated governance escalation for unacceptable risks.*
REQ-MOD-RSK-006	SHOULD	The Risk Module SHOULD provide a risk heat map visualization (likelihood × impact grid) exportable for board and committee reporting.	*Standard executive and board risk reporting format.*

5.2.10 Security, Privacy & Audit (MOD-36)

Req ID	Priority	Requirement Statement	Rationale
REQ-MOD-SEC-001	SHALL	The Security Module SHALL manage identity and access: user identities, authentication configuration, session management, and MFA enrollment.	*Centralized identity governance.*
REQ-MOD-SEC-002	SHALL	The Security Module SHALL maintain a vulnerability register: CVEs linked to CMDB CIs, severity ratings, patch status, and SLA tracking.	*Structured vulnerability management lifecycle.*
REQ-MOD-SEC-003	SHALL	The Security Module SHALL manage security incidents: creation, triage, classification (P1–P4), SLA timers, investigation, containment, remediation, and PIR.	*Full security incident response lifecycle.*
REQ-MOD-SEC-004	SHALL	The Security Module SHALL provide a privacy management sub-module: Records of Processing Activities (RoPA), DPIA management, consent management, and DSAR workflow.	*GDPR, CCPA, and equivalent privacy law compliance.*
REQ-MOD-SEC-005	SHALL	The Security Module SHALL provide a cryptographic vault interface for key management; raw key material SHALL NOT be accessible outside the vault interface.	*HSM-compatible key management architecture.*
REQ-MOD-SEC-006	SHALL	The Security Module SHALL enforce MFA for all privileged operations (system admin, RBAC changes, key management) regardless of user-level MFA enrollment status.	*Privilege escalation attack mitigation.*
REQ-MOD-SEC-007	SHOULD	The Security Module SHOULD integrate with external SIEM systems by publishing all security events to a configurable webhook or event bus topic.	*Enterprise SOC integration.*

5.2.11 IT & Asset Management (MOD-18)

Req ID	Priority	Requirement Statement	Rationale
REQ-MOD-IT-001	SHALL	The IT Module SHALL maintain a Configuration Management Database (CMDB) tracking all IT assets as Configuration Items (CIs) with: CI type, owner, status, location, and relationship map.	*ITIL-compliant asset and configuration management.*
REQ-MOD-IT-002	SHALL	The IT Module SHALL manage the IT service management lifecycle: incidents, service requests, problems, and changes with SLA tracking and escalation.	*ITIL ITSM process compliance.*
REQ-MOD-IT-003	SHALL	The IT Module SHALL enforce change management: changes SHALL be classified (standard, normal, emergency); normal and emergency changes SHALL require CAB or emergency approver authorization.	*Prevents unauthorized changes to production systems.*
REQ-MOD-IT-004	SHALL	The IT Module SHALL track software license entitlements vs deployed instances; over-deployment SHALL trigger a compliance alert.	*Software license compliance management.*
REQ-MOD-IT-005	SHOULD	The IT Module SHOULD support automated CMDB discovery integration via configuration scan agents or SNMP/API integrations.	*Reduces manual CMDB maintenance burden.*

5.2.12 Marketing System — Soko (MOD-22)

Req ID	Priority	Requirement Statement	Rationale
REQ-MOD-MKT-001	SHALL	Soko SHALL manage marketing campaigns with: audience definition, channel strategy, budget, timeline, and KPI targets.	*Full campaign planning and management.*
REQ-MOD-MKT-002	SHALL	Soko SHALL execute strategy pack evaluation on campaign creation and signal ingestion; a minimum of 70 built-in strategy packs SHALL be available on initialization.	*AI-assisted campaign strategy is a core Soko capability.*
REQ-MOD-MKT-003	SHALL	Soko SHALL ingest market signals in real time and re-evaluate campaign insights as signals accumulate; updated insights SHALL be surfaced to the marketing manager.	*Dynamic campaign optimization as signals arrive.*
REQ-MOD-MKT-004	SHALL	Soko SHALL track campaign performance KPIs: CAC, conversion rate, click-through rate, revenue attributed, and ROI; KPIs SHALL update in near-real-time.	*Campaign performance visibility.*
REQ-MOD-MKT-005	SHOULD	Soko SHOULD integrate with the CRM Module to pass marketing-qualified leads with attribution data when configured conversion thresholds are met.	*Closed-loop marketing-to-sales attribution.*

5.2.13 Sales Management (MOD-34)

Req ID	Priority	Requirement Statement	Rationale
REQ-MOD-SAL-001	SHALL	The Sales Module SHALL manage the sales opportunity lifecycle: lead, qualify, propose, negotiate, closed-won, closed-lost with stage-gate criteria.	*Structured pipeline management.*
REQ-MOD-SAL-002	SHALL	The Sales Module SHALL compute weighted pipeline value, win rate, average sales cycle, and forecast accuracy as standard KPIs.	*Core sales management metrics.*

REQ-MOD-SAL-003	SHALL	The Sales Module SHALL support sales territory management: assignment of accounts and leads to representatives by territory, vertical, or named account model.	*Territory governance and coverage management.*
REQ-MOD-SAL-004	SHALL	The Sales Module SHALL compute sales commissions based on configurable commission plans; commission statements SHALL be generated per representative per period.	*Incentive compensation management.*
REQ-MOD-SAL-005	SHOULD	The Sales Module SHOULD provide AI-assisted deal scoring: probability-to-close predictions based on historical win/loss patterns and deal attributes.	*Improves forecast accuracy.*

5.2.14 Supply Chain, Inventory & Logistics (MOD-37)

Req ID	Priority	Requirement Statement	Rationale
REQ-MOD-SC-001	SHALL	The Supply Chain Module SHALL manage the procurement lifecycle: requisition, RFQ, supplier selection, purchase order, goods receipt, and 3-way match.	*End-to-end procurement management.*
REQ-MOD-SC-002	SHALL	The Supply Chain Module SHALL maintain an inventory ledger with real-time stock levels, reorder points, safety stock, and EOQ calculation.	*Inventory management with automatic replenishment triggers.*
REQ-MOD-SC-003	SHALL	The Supply Chain Module SHALL emit sc.inventory.reorder_triggered when stock falls to or below the reorder point; the event SHALL include item, location, and suggested order quantity.	*Automated procurement trigger.*
REQ-MOD-SC-004	SHALL	The Supply Chain Module SHALL maintain a qualified supplier registry with: contact, payment terms, delivery lead time, quality rating, and ESG score.	*Supplier governance and performance tracking.*
REQ-MOD-SC-005	SHOULD	The Supply Chain Module SHOULD support multi-echelon inventory across multiple warehouse locations with transfer order management between locations.	*Multi-site inventory management.*

5.2.15 Remaining Modules (MOD-05 through MOD-42, excluding those above)

The following modules share the same universal contract (REQ-MOD-001 through REQ-MOD-012) and have the specific functional requirements below:

Module	ID	Key Functional Requirements
Business Development & Analysis	MOD-05	Pipeline management for partnerships, grants, and business opportunities. Opportunity scoring and conversion tracking. Market intelligence repository.
Enterprise Content Management	MOD-06	Document lifecycle with draft/review/publish states. Version history with diff view. Full-text search indexing. Document access control at asset level.
Communications	MOD-07	Multi-channel message dispatch: email, SMS, push, in-app. Template rendering via MOD-39. Delivery status tracking. Opt-out/unsubscribe management. Bulk and transactional send.
CRM — Relationships	MOD-08	Stakeholder lifecycle: lead, prospect, client, partner. Interaction history log. Relationship health scoring. 360-degree view linking to Finance (invoices), Sales (opportunities), and Legal (contracts).

Organization Design System	MOD-09	Brand identity registry: colors, fonts, logos, usage rules. UI component library. Design token management. Asset export in SVG/PNG/PDF formats.
Engineering & Technology	MOD-10	Architecture Decision Record (ADR) management. Technical debt register with aging. Incident post-mortem library. Engineering velocity metrics from Work Module.
ESG, CSR & Sustainability	MOD-11	Environmental KPI tracking: Scope 1/2/3 emissions, water, waste. Social metrics: diversity, community investment, workforce wellbeing. Governance scores. GRI/SASB/TCFD reporting templates.
Enterprise Management & Admin	MOD-12	Module configuration and health dashboard for system operators. Cross-module dependency visualization. System-wide configuration change management with approval workflow.
Investment Management	MOD-17	Investment pipeline from screening to deployment. Appraisal models: DCF, IRR, MOIC, DPI. Portfolio company performance monitoring. LP capital account and distribution management.
Knowledge Management	MOD-19	Article authoring with rich text and media. Taxonomy and tag management. Related article linking. Full-text search via search driver. Version history. Article usage analytics.
Learning & Development	MOD-20	Learning path design with prerequisite chains. SCORM/xAPI content compatibility. Assessment and quiz engine with configurable pass marks. CPD credit tracking. Certificate generation.
Management & Strategy	MOD-21	OKR hierarchy: organizational → department → individual. PESTLE and SWOT analysis tools. Strategic initiative management with health tracking. OKR progress roll-up to executive dashboard.
Master Data Management	MOD-23	Golden record management for entities, products, and persons. Deduplication workflow. Data quality scoring. Cross-module reference data synchronization. Lineage tracking.
Office & Facilities	MOD-24	Space inventory with floor plan references. Desk/room booking with calendar integration. Asset (furniture, equipment) assignment. Visitor management. Facilities maintenance request workflow.
Operations Management	MOD-25	Operational process SLA definition and monitoring. Throughput, cycle time, and quality KPIs per process. Exception and escalation management. Capacity planning model.
Portal, Hub & Dashboard	MOD-26	Personalized user landing page with role-appropriate widgets. Global search across all modules. Notification center (all module notifications aggregated). Quick-action shortcuts.
Portfolio & Program Management	MOD-27	Portfolio hierarchy: strategy → program → project → work package. Portfolio health roll-up (RAG status). Resource demand vs supply across portfolio. Stage gate and approval gates.
PR & Branding	MOD-28	Press release lifecycle and distribution. Media contact registry. Brand mention monitoring. Crisis communications playbook management. Share of voice tracking.
Process, Orchestration & Workflow	MOD-29	Process definition with BPMN-compatible flow logic. Workflow template library. Process instance execution with state machine. SLA tracking per process step. DLQ for failed process instances.
Product, Services & Solutions	MOD-30	Product catalog with versioning. Product roadmap with milestone and release planning. Feature backlog with priority scoring. Product KPIs: adoption, revenue, NPS, defect rate.
Production, Manufacturing & Fabrication	MOD-31	Bill of Materials (BOM) management with multi-level explosion. Work order lifecycle. Production scheduling with capacity constraints. Quality inspection and NCR management.
Requirements Management	MOD-32	Requirements hierarchy: business → functional → non-functional. Traceability matrix: requirements to test cases to deliverables. Change impact analysis. Stakeholder sign-off tracking.

Schedule Management	MOD-35	Enterprise calendar with multi-timezone support. Event types: meetings, milestones, deadlines, blackouts. Resource availability tracking. Calendar export in iCal format. Conflict detection.
Logistics & Warehouse	MOD-37	Warehouse location management with bin/rack/zone structure (shared with Supply Chain). Inbound and outbound shipment tracking. Pick/pack/ship workflow. Carrier integration hooks.
Team & Cooperative Management	MOD-38	Team registry with roles and membership. Team OKRs and performance tracking. Retrospective tooling with action item creation. Team health and engagement metrics.
Templating System	MOD-39	Template library with variable substitution and conditional blocks. Brand-consistent output in DOCX, PDF, and HTML. Template versioning. Template approval workflow.
Work Management	MOD-40	Work item types: task, story, bug, epic, work order. Board and list views. Sprint/cycle management. Time tracking against items. Dependency linking. Velocity and burndown metrics.
Custom UME Modules	MOD-41	Namespace: custom.ume.{vendor}.{module_name}. Full KernelModule trait compliance. SDK scaffold generation. Marketplace metadata for discovery. Module signing for distribution.
Custom Organization Modules	MOD-42	Namespace: org.custom.{vendor}.*. Full KernelModule trait compliance. Organization-private deployment. No cross-organization data sharing via this namespace.

6. Interface & API Requirements

UME exposes multiple interface layers: the HTTP REST API (ume_server), the CLI runtime (ume_runtime), the Java client suite, and the SDK for module development. Requirements for each are defined here.

6.1 HTTP REST API (ume_server)

Req ID	Priority	Requirement Statement	Rationale
REQ-API-001	SHALL	The HTTP API SHALL follow REST semantics: resources identified by URLs, standard HTTP verbs (GET, POST, PUT, PATCH, DELETE), and HTTP status codes per RFC 7231.	*Interoperability with any HTTP client or integration platform.*
REQ-API-002	SHALL	All API endpoints SHALL require authentication; unauthenticated requests SHALL receive HTTP 401; requests with valid authentication but insufficient permissions SHALL receive HTTP 403.	*API access must be authenticated and authorized.*
REQ-API-003	SHALL	The HTTP API SHALL implement rate limiting per authenticated client; configurable thresholds; HTTP 429 SHALL be returned with Retry-After header on limit breach.	*Protects system from runaway clients and abuse.*
REQ-API-004	SHALL	All API requests and responses SHALL use JSON (application/json); datetime fields SHALL use ISO 8601 UTC format (RFC3339).	*Consistent data serialization across all clients.*
REQ-API-005	SHALL	The HTTP API SHALL support pagination for all list endpoints using cursor-based pagination; page size SHALL be configurable; default page size SHALL be 50.	*Prevents unbounded data transfers for large collections.*
REQ-API-006	SHALL	The HTTP API SHALL support filtering, sorting, and field selection (sparse fieldsets) for all list endpoints via query parameters.	*Enables efficient, targeted API queries.*
REQ-API-007	SHALL	The HTTP API SHALL return structured error responses with: error_code, message, detail (array of field-level errors), request_id, and documentation_url.	*Machine-readable errors enable robust client error handling.*
REQ-API-008	SHALL	The HTTP API SHALL expose a /health endpoint returning kernel health, individual module health states, and driver health in a single JSON response.	*Required for load balancer and orchestrator health probes.*
REQ-API-009	SHALL	The HTTP API SHALL expose a /metrics endpoint in Prometheus text format exposing all kernel and module metrics.	*Standard observability integration.*
REQ-API-010	SHALL	The HTTP API SHALL support TLS 1.2+ for all production deployments; HTTP-only SHALL be disabled by default in production mode.	*Transport security requirement.*
REQ-API-011	SHOULD	The HTTP API SHOULD provide an OpenAPI 3.1 specification document at /openapi.json; the specification SHALL be auto-generated from route definitions and kept in sync.	*Enables SDK generation and developer self-service.*
REQ-API-012	SHOULD	The HTTP API SHOULD support WebSocket connections for real-time event streaming; clients SHALL be able to subscribe to event bus topics via a WebSocket endpoint.	*Enables live dashboard updates without polling.*
REQ-API-013	SHOULD	The HTTP API SHOULD support bulk operation endpoints (batch create, batch update) to reduce round-trip overhead for high-volume data import scenarios.	*Operational efficiency for data migration and integration.*

REQ-API-014	MAY	The HTTP API MAY support GraphQL as an alternative query interface for read operations, generated from the OpenAPI schema.	*Advanced query flexibility for complex client applications.*

6.2 CLI Runtime (ume_runtime)

Req ID	Priority	Requirement Statement	Rationale
REQ-API-020	SHALL	The ume_runtime CLI SHALL support an interactive REPL mode and a non-interactive batch mode accepting commands from stdin or script files.	*Supports both interactive admin use and scripted automation.*
REQ-API-021	SHALL	The ume_runtime SHALL provide commands for: kernel management (start, stop, status, config), module management (list, enable, disable, health), and system management (backup, restore, audit-query).	*Complete system management without requiring a GUI.*
REQ-API-022	SHALL	The CLI SHALL support structured output in JSON and human-readable table formats; format selection SHALL be via --format flag.	*Enables both human use and script parsing.*
REQ-API-023	SHALL	CLI commands that mutate system state SHALL require explicit confirmation or a --yes flag to prevent accidental execution.	*Prevents accidental destructive operations.*
REQ-API-024	SHOULD	The CLI SHOULD provide shell tab-completion for commands, flags, and common argument values.	*Improves operator productivity.*

6.3 Java Client Suite

Req ID	Priority	Requirement Statement	Rationale
REQ-API-030	SHALL	The Java client SHALL provide GUI, shell (interactive REPL), and watch (live dashboard) modes selectable at launch.	*Multiple interaction patterns for different user contexts.*
REQ-API-031	SHALL	The Java GUI client SHALL implement the full module set as navigable screens with consistent UX patterns: list, detail, form, and dashboard views.	*Full-featured GUI for non-technical users.*
REQ-API-032	SHALL	The Java client SHALL authenticate via the HTTP API using JWT bearer tokens; token refresh SHALL be handled transparently without user interruption.	*Seamless authenticated session management.*
REQ-API-033	SHALL	The Java shell client SHALL support all ume_runtime CLI commands via an interactive menu-driven interface.	*CLI capability parity in a guided UI.*
REQ-API-034	SHALL	The Java watch client SHALL display configurable real-time dashboards refreshed from the WebSocket event stream; panels SHALL be user-configurable.	*Live monitoring without manual refresh.*
REQ-API-035	SHOULD	The Java client SHALL support deep-linking: a URL scheme (ume://) SHALL allow direct navigation to any module resource.	*Enables cross-application linking and notification deep links.*
REQ-API-036	SHOULD	The Java client SHOULD support offline read-only mode with locally cached data; sync SHALL resume automatically on connectivity restoration.	*Usability in low-connectivity environments.*

6.4 Module Development SDK

Req ID	Priority	Requirement Statement	Rationale
REQ-API-040	SHALL	The UME Module SDK SHALL provide complete Rust trait definitions and mock implementations for all kernel facades: KernelModule, RbacEngine, EventBus, StorageManager, LogAuditManager, and all driver types.	*Developers can build and test modules without a running kernel.*
REQ-API-041	SHALL	The SDK SHALL provide a scaffold generator (tools/generate_custom_module) that creates a complete, compilable module boilerplate from a module name and domain.	*Reduces module bootstrap time from days to minutes.*
REQ-API-042	SHALL	The SDK SHALL include integration test harness utilities: MockKernel, MockStorage, and EventCapture for assertion-based event testing.	*Enables thorough, fast module testing.*
REQ-API-043	SHALL	The SDK SHALL be versioned independently and maintain backward compatibility: a module compiled against SDK N SHALL run on kernel versions N and N+1.	*Prevents module breakage on kernel upgrades.*
REQ-API-044	SHOULD	The SDK SHOULD include a schema validator tool that verifies module data models comply with ume_core schema conventions before compilation.	*Catches schema violations at build time.*
REQ-API-045	MAY	The SDK MAY provide language bindings for Python and TypeScript to enable module development in those languages via FFI.	*Broadens the developer ecosystem.*

7. Data Architecture Requirements

7.1 Data Model & Schema

Req ID	Priority	Requirement Statement	Rationale
REQ-DAT-001	SHALL	All persistent domain data models SHALL be defined as types in the ume_core crate; no module-private persistence schemas are permitted for data shared across module boundaries.	*Single schema definition prevents cross-module data inconsistency.*
REQ-DAT-002	SHALL	All entity IDs SHALL be UUID v4 generated by the creating module; sequential integer IDs SHALL NOT be used as public identifiers.	*UUIDs prevent enumeration attacks and support distributed generation.*
REQ-DAT-003	SHALL	All timestamped fields SHALL store UTC timestamps in RFC3339 format; local time presentation is exclusively a client responsibility.	*Consistent temporal data across multi-timezone organizations.*
REQ-DAT-004	SHALL	All domain entities SHALL carry: id (UUID v4), created_at (UTC), updated_at (UTC), created_by (subject_id), and version (optimistic lock counter).	*Universal audit fields on every domain entity.*
REQ-DAT-005	SHALL	Optimistic concurrency control SHALL be applied to all mutating operations on entities with a version field; stale-version updates SHALL be rejected with HTTP 409 Conflict.	*Prevents lost updates under concurrent access.*
REQ-DAT-006	SHALL	Soft-delete SHALL be used for all business entities; hard-delete SHALL be reserved for data subject erasure requests (GDPR Art. 17) and SHALL be audited.	*Preserves referential integrity and audit trail.*
REQ-DAT-007	SHOULD	Complex domain entities SHOULD support JSON Schema validation at the API boundary before persistence; validation errors SHALL be returned in structured error format.	*Prevents invalid data from reaching the storage layer.*

7.2 Multi-Tenancy & Isolation

Req ID	Priority	Requirement Statement	Rationale
REQ-DAT-010	SHALL	The UME platform SHALL support multi-organization deployment; each organization's data SHALL be logically isolated via organization_id partitioning on all data tables.	*Enables SaaS deployment while maintaining data isolation.*
REQ-DAT-011	SHALL	Cross-organization data access SHALL be architecturally impossible at the data layer: all queries SHALL be scoped by organization_id without exception.	*No accidental data leakage between tenants.*
REQ-DAT-012	SHALL	Multi-entity organizations SHALL be represented as entity records within a single organization; cross-entity access within the same organization SHALL be RBAC-controlled.	*Supports holding companies without requiring separate org deployments.*
REQ-DAT-013	SHOULD	The platform SHOULD support physical database isolation for high-security or regulatory tenants (own database schema or database instance).	*Meets data residency and sovereignty requirements for regulated tenants.*

7.3 Data Integrity & Consistency

Req ID	Priority	Requirement Statement	Rationale
REQ-DAT-020	SHALL	All writes to the primary storage backend SHALL use transactions with serializable isolation for operations involving multiple table mutations.	*Prevents partial writes that leave data in inconsistent state.*
REQ-DAT-021	SHALL	Foreign key relationships SHALL be enforced at the database level for all relational backends (PostgreSQL, SQLite); soft constraints MAY be used for cross-module references.	*Database-level integrity prevents orphaned references.*
REQ-DAT-022	SHALL	The system SHALL maintain eventual consistency semantics for analytics aggregates and derived data; a maximum staleness of 60 seconds is acceptable for non-real-time KPIs.	*Clearly defined consistency boundary for analytics data.*
REQ-DAT-023	SHALL	Financial data (journal entries, invoices, payments) SHALL be stored with decimal precision of minimum 2 decimal places for functional currency; exchange rate data SHALL use 6 decimal places.	*Prevents rounding errors in financial calculations.*
REQ-DAT-024	SHOULD	The system SHOULD support event sourcing for high-value audit domains (Finance, GRC, Board); state SHALL be reconstructable from the event log.	*Strongest possible audit trail for regulated domains.*

7.4 Data Migration & Evolution

Req ID	Priority	Requirement Statement	Rationale
REQ-DAT-030	SHALL	All schema changes SHALL be applied via versioned migration scripts managed by the StorageManager; migrations SHALL be idempotent and forward-only.	*Safe schema evolution with zero manual intervention.*
REQ-DAT-031	SHALL	Migration scripts SHALL be tested against a copy of the production schema in CI before deployment; failed migrations SHALL automatically roll back.	*Prevents migration failures from impacting production.*
REQ-DAT-032	SHALL	Breaking schema changes (column removal, type changes) SHALL be preceded by a deprecation migration period of at least one major version.	*Prevents data loss from premature schema breaking changes.*
REQ-DAT-033	SHOULD	The platform SHOULD provide an import/export API for each module's data in standard formats (CSV, JSON) to support data migration between UME instances.	*Enables organizational data portability.*

7.5 Master Data Management

Req ID	Priority	Requirement Statement	Rationale
REQ-DAT-040	SHALL	The MDM Module (MOD-23) SHALL serve as the golden record authority for shared reference data: organizations, persons, products, and geographic data.	*Prevents reference data fragmentation across modules.*
REQ-DAT-041	SHALL	All modules consuming MDM data SHALL subscribe to MDM change events and refresh their local reference cache within the configured staleness window.	*Keeps module-local reference data current.*
REQ-DAT-042	SHALL	The MDM Module SHALL implement a deduplication workflow: potential duplicate records SHALL be surfaced for	*Maintains golden record quality over time.*

		human review and merge; merges SHALL be audited with lineage preserved.	

8. Security Requirements

8.1 Authentication

Req ID	Priority	Requirement Statement	Rationale
REQ-SEC-001	SHALL	The system SHALL support multiple authentication mechanisms: username/password (bcrypt hashed), OAuth2/OIDC, SAML 2.0, and API key authentication.	*Accommodates all common enterprise identity configurations.*
REQ-SEC-002	SHALL	Passwords SHALL be hashed with bcrypt (minimum cost factor 12) or Argon2id; plaintext passwords SHALL never be stored or logged.	*Industry-standard password storage security.*
REQ-SEC-003	SHALL	The system SHALL support Multi-Factor Authentication (MFA); TOTP (RFC 6238) and FIDO2/WebAuthn SHALL be supported as MFA factors.	*Protects against credential compromise.*
REQ-SEC-004	SHALL	MFA SHALL be mandatory for system administrators and users with RBAC write permissions; enforcement SHALL not be bypassable via configuration.	*Privileged account protection cannot be weakened.*
REQ-SEC-005	SHALL	Session tokens SHALL be JWT (RS256 signed); token expiry SHALL default to 60 minutes for access tokens and 7 days for refresh tokens; both SHALL be configurable.	*Limits the window of exposure for compromised tokens.*
REQ-SEC-006	SHALL	The system SHALL implement account lockout after configurable consecutive failed authentication attempts; default 5 attempts with 15-minute lockout.	*Brute force attack mitigation.*
REQ-SEC-007	SHALL	All authentication events (success, failure, lockout, MFA challenge) SHALL produce audit records.	*Non-repudiation for all authentication events.*
REQ-SEC-008	SHOULD	The system SHOULD support IP allowlisting for API access per organization; requests from non-allowlisted IPs SHALL receive HTTP 403.	*Network-level access restriction for high-security deployments.*

8.2 Authorization

Req ID	Priority	Requirement Statement	Rationale
REQ-SEC-010	SHALL	The RBAC engine SHALL be the sole authorization decision point; modules SHALL NOT implement alternative authorization mechanisms.	*Consistent, centrally auditable authorization.*
REQ-SEC-011	SHALL	Permission checks SHALL occur on every mutating operation regardless of the operation's origin (API, event handler, scheduled job, CLI).	*No authorization bypass paths.*
REQ-SEC-012	SHALL	The principle of least privilege SHALL be the default: new users SHALL have no permissions until explicitly granted; no default wildcard grants.	*Prevents accidental over-provisioning.*
REQ-SEC-013	SHALL	Separation of duties constraints SHALL be configurable: specific permission combinations (e.g., payment.approve + payment.create) SHALL be blockable for the same subject.	*Fraud prevention for financial and approval workflows.*
REQ-SEC-014	SHALL	RBAC permission grants and revocations SHALL require a second authorizer for permissions above a configured sensitivity level.	*Four-eyes principle for privileged access changes.*

8.3 Data Protection

Req ID	Priority	Requirement Statement	Rationale
REQ-SEC-020	SHALL	All data at rest SHALL be encrypted using AES-256-GCM; encryption SHALL be applied at the storage layer by the StorageDriver.	*Data-at-rest protection for all persistent data.*
REQ-SEC-021	SHALL	All data in transit SHALL use TLS 1.2 or higher; TLS 1.0 and 1.1 SHALL be disabled; cipher suites SHALL follow OWASP TLS recommendations.	*Transport security across all interfaces.*
REQ-SEC-022	SHALL	Encryption keys SHALL be managed via the Security Module vault interface; keys SHALL be rotatable without system downtime.	*Key lifecycle management without availability impact.*
REQ-SEC-023	SHALL	Personally Identifiable Information (PII) fields SHALL be tagged in the schema and SHALL support field-level encryption as a configurable option.	*GDPR and privacy-by-design requirement.*
REQ-SEC-024	SHALL	Sensitive data fields (passwords, API keys, tokens, PII) SHALL be masked or redacted in all log output regardless of log level.	*Prevents accidental credential and PII exposure in logs.*
REQ-SEC-025	SHALL	Backup artifacts SHALL be encrypted with AES-256; backup encryption keys SHALL be stored separately from backup artifacts.	*Data protection for backup storage.*

8.4 Input Validation & Injection Prevention

Req ID	Priority	Requirement Statement	Rationale
REQ-SEC-030	SHALL	All API inputs SHALL be validated against the declared schema before processing; inputs failing validation SHALL be rejected with HTTP 422 and structured error detail.	*First-line defense against malformed and malicious inputs.*
REQ-SEC-031	SHALL	All database queries SHALL use parameterized queries or prepared statements; string concatenation in SQL queries is prohibited.	*SQL injection prevention.*
REQ-SEC-032	SHALL	All user-supplied content rendered in HTML contexts SHALL be HTML-escaped; no raw user content SHALL be rendered as HTML without sanitization.	*XSS prevention.*
REQ-SEC-033	SHALL	File uploads SHALL be validated for MIME type, file extension, and maximum size; uploaded files SHALL be stored outside the web root.	*File upload attack prevention.*
REQ-SEC-034	SHOULD	The system SHOULD implement Content Security Policy (CSP) headers on all web responses to restrict script sources.	*Defense-in-depth against XSS attacks.*

8.5 Security Monitoring & Response

Req ID	Priority	Requirement Statement	Rationale

REQ-SEC-040	SHALL	The system SHALL produce security events for: authentication failures, RBAC denials, privilege escalations, unusual access patterns, and system configuration changes.	*SIEM integration and SOC monitoring.*
REQ-SEC-041	SHALL	The Security Module SHALL support automated security incident creation from configurable rule-based detections on the event stream.	*Automated threat detection.*
REQ-SEC-042	SHALL	The system SHALL support vulnerability scanning integration; CVEs matched to CMDB CIs SHALL automatically create vulnerability records in the IT Module.	*Automated vulnerability management.*
REQ-SEC-043	SHALL	A data breach detection and notification workflow SHALL be available; on activation, it SHALL guide through regulatory notification timelines (GDPR 72-hour deadline).	*Regulatory compliance for data breach notification.*
REQ-SEC-044	SHOULD	The system SHOULD generate automated dependency vulnerability reports (SBOM with CVE cross-reference) for the UME platform itself on each release.	*Supply chain security transparency.*

9. Non-Functional Requirements

9.1 Performance

NFR ID	Attribute	Target	Condition / Notes
REQ-NFR-001	API Response Time (P50)	< 100 ms	Standard read operations; single entity lookup; warm cache
REQ-NFR-002	API Response Time (P95)	< 500 ms	Standard read and write operations under normal load
REQ-NFR-003	API Response Time (P99)	< 2000 ms	All operations including analytics queries; under peak load
REQ-NFR-004	Dashboard KPI Refresh	< 60 seconds	Time from domain event to dashboard metric update
REQ-NFR-005	Batch Report Generation	< 30 seconds	Financial statements, board packs; 12-month dataset
REQ-NFR-006	Module Health Check	< 500 ms	health_check() response time; timeout triggers supervisor action
REQ-NFR-007	Kernel Boot Time	< 30 seconds	Full kernel + all modules initialized and in Running state
REQ-NFR-008	Search Query Response	< 1 second	Full-text search across Knowledge and CMS modules; 1M+ documents
REQ-NFR-009	Event Bus Throughput	> 10,000 events/sec	Sustained in-memory bus throughput; single-node deployment
REQ-NFR-010	Audit Record Write	< 50 ms	Audit record persistence; SHALL NOT block the calling operation

9.2 Scalability

Req ID	Priority	Requirement Statement	Rationale
REQ-NFR-020	SHALL	The system SHALL support horizontal scaling of the HTTP API tier via stateless server instances behind a load balancer; no session state SHALL be stored in the application tier.	*Enables linear capacity scaling with instance count.*
REQ-NFR-021	SHALL	The system SHALL support a minimum of 500 concurrent authenticated API connections per server instance without performance degradation below P95 targets.	*Baseline concurrent user capacity.*
REQ-NFR-022	SHALL	The system SHALL support organizations with up to 100,000 employees, 1,000,000 work items, 10,000,000 journal entries, and 50,000 active CRM contacts within a single deployment.	*Sufficient scale for large enterprise deployment.*
REQ-NFR-023	SHALL	The database tier SHALL support read replica configuration; the StorageManager SHALL route read operations to replicas when configured.	*Database scalability for read-heavy workloads.*
REQ-NFR-024	SHOULD	The system SHOULD support auto-scaling via Kubernetes HPA based on CPU and custom event queue depth metrics.	*Cloud-native elastic scaling.*

REQ-NFR-025	SHOULD	Analytics queries over large datasets SHOULD be executed asynchronously with result caching; re-queries on cached results SHALL return within 1 second.	*Prevents analytics queries from impacting transactional performance.*

9.3 Availability & Reliability

NFR ID	Attribute	Target	Measurement Period
REQ-NFR-030	Service Availability (Production)	≥ 99.9%	Rolling 30-day window; excludes planned maintenance windows
REQ-NFR-031	Service Availability (Enterprise SLA)	≥ 99.95%	Rolling 30-day window; premium enterprise tier
REQ-NFR-032	Recovery Time Objective (RTO)	< 4 hours	Full system recovery from complete failure with backup restore
REQ-NFR-033	Recovery Point Objective (RPO)	< 1 hour	Maximum data loss on unplanned failure with default backup config
REQ-NFR-034	Planned Maintenance Window	< 2 hours/month	With advance notice of 72 hours to users
REQ-NFR-035	Mean Time to Detect (MTTD)	< 5 minutes	Automated health check detects and alerts on module failure
REQ-NFR-036	Mean Time to Recover (MTTR)	< 30 minutes	Automated supervisor recovery for transient module failures
REQ-NFR-037	Data Durability	≥ 99.999999%	Annual probability of data loss with configured backup strategy

9.4 Observability

Req ID	Priority	Requirement Statement	Rationale
REQ-NFR-040	SHALL	The system SHALL expose metrics in Prometheus format covering: kernel health, per-module operation throughput and latency, executor queue depths, memory region utilization, and storage latency.	*Full-stack observability without proprietary monitoring lock-in.*
REQ-NFR-041	SHALL	All log entries SHALL be structured (JSON) and SHALL include correlation_id for distributed request tracing; logs SHALL be emittable to stdout, file, and syslog sinks.	*Enables log aggregation in ELK, Splunk, and similar platforms.*
REQ-NFR-042	SHALL	The system SHALL support distributed tracing via OpenTelemetry (OTLP); trace spans SHALL cover API request handling, module calls, storage operations, and event bus publish/subscribe.	*End-to-end request tracing across all system layers.*
REQ-NFR-043	SHALL	A system status page SHALL be exposed showing: kernel state, module health matrix, driver health, active alert count, and last 24h availability.	*Operational transparency for operators and users.*
REQ-NFR-044	SHOULD	The system SHOULD provide pre-built dashboards for Grafana and Kibana covering kernel, module, API, and infrastructure metrics.	*Reduces observability setup time for operators.*

9.5 Usability

Req ID	Priority	Requirement Statement	Rationale
REQ-NFR-050	SHALL	The portal and Java GUI SHALL achieve a minimum System Usability Scale (SUS) score of 75 (Good) in usability testing with target user groups.	*UME must be genuinely usable without training for its primary workflows.*
REQ-NFR-051	SHALL	All error messages presented to end users SHALL be written in plain language and SHALL include suggested corrective actions; raw system error codes SHALL NOT be displayed.	*User experience and support deflection.*
REQ-NFR-052	SHALL	Time-to-complete for the top 10 most frequent user workflows (defined in UME-WUC-001/002) SHALL be measurable and SHALL not increase between major releases.	*Regression protection for key workflow efficiency.*
REQ-NFR-053	SHOULD	The portal SHALL be accessible to WCAG 2.1 Level AA standards; accessibility testing SHALL be included in the release checklist.	*Inclusivity and accessibility compliance.*
REQ-NFR-054	SHOULD	The portal SHALL support responsive design for tablet (768px+) and desktop (1024px+) screen sizes; mobile-friendly view for notification center and dashboards.	*Multi-device access for modern workplace patterns.*

9.6 Maintainability

Req ID	Priority	Requirement Statement	Rationale
REQ-NFR-060	SHALL	The codebase SHALL maintain a minimum test coverage of 80% line coverage for all ume_core and kernel crates; module crates SHALL maintain minimum 70% coverage.	*Baseline regression protection.*
REQ-NFR-061	SHALL	All public API surfaces SHALL have corresponding integration tests that run against a live kernel instance; these tests SHALL run in CI on every pull request.	*API contract regression detection.*
REQ-NFR-062	SHALL	The system SHALL support zero-downtime deployment via rolling upgrades: new version instances SHALL start accepting traffic before old instances are terminated.	*Continuous delivery without maintenance windows.*
REQ-NFR-063	SHALL	Every breaking change to a public API or module SDK interface SHALL be versioned; the old interface SHALL be deprecated with a minimum 1-minor-version support window.	*Protects integrators from unannounced breaking changes.*
REQ-NFR-064	SHOULD	The Rust codebase SHOULD pass clippy at the warn level with no allowed warnings in CI; all new code SHALL pass cargo fmt validation.	*Consistent code quality and style.*
REQ-NFR-065	SHOULD	Dependency upgrades SHALL be automated via Dependabot or equivalent; security-relevant dependency upgrades SHALL be applied within 72 hours of CVE publication.	*Supply chain security maintenance.*

10. Deployment & Infrastructure Requirements

10.1 Deployment Models

UME SHALL support three primary deployment models, each with defined infrastructure requirements:

Model	Description	Minimum Infrastructure	Target Org Scale
Embedded / Single-Process	Kernel + modules + SQLite storage in one binary. Ideal for solo operators and small orgs.	1 CPU core, 512 MB RAM, 10 GB disk	1–20 users
Standard / Containerized	Docker Compose: ume_server + PostgreSQL + Redis. Suitable for most organizations.	2 CPU cores, 4 GB RAM, 100 GB SSD	20–500 users
Enterprise / Kubernetes	Helm chart: multiple ume_server replicas + managed PostgreSQL (RDS/Cloud SQL) + Redis cluster.	8+ CPU cores, 16 GB RAM, 500 GB SSD	500–100,000 users

Req ID	Priority	Requirement Statement	Rationale
REQ-DEP-001	SHALL	The system SHALL be packaged as a single statically-linked Rust binary (ume_server) with no runtime dependencies other than the OS libc.	*Simplifies deployment and eliminates runtime dependency conflicts.*
REQ-DEP-002	SHALL	The system SHALL provide an official Docker image published to a container registry; images SHALL be built for linux/amd64 and linux/arm64 architectures.	*Supports cloud and ARM-based infrastructure.*
REQ-DEP-003	SHALL	The system SHALL provide a Helm chart for Kubernetes deployment with configurable values for all environment-specific settings.	*Standard Kubernetes deployment mechanism.*
REQ-DEP-004	SHALL	All environment-specific configuration SHALL be injectable via environment variables; no hardcoded configuration SHALL exist in any deployment artifact.	*12-factor app compliance for cloud-native deployment.*
REQ-DEP-005	SHALL	The system SHALL perform database schema migrations automatically on startup; migrations SHALL be idempotent and safe to run on a running system with traffic.	*Zero-friction upgrades without manual migration steps.*
REQ-DEP-006	SHALL	Rolling deployment SHALL be supported: the system SHALL handle requests with mixed old/new schema versions for a window of at least one minor version.	*Zero-downtime rolling upgrades.*
REQ-DEP-007	SHOULD	The system SHOULD provide Terraform and/or Pulumi infrastructure-as-code modules for common cloud targets (AWS, GCP, Azure).	*Repeatable, auditable infrastructure provisioning.*
REQ-DEP-008	SHOULD	The system SHOULD provide a one-command local development setup (docker-compose up or cargo run) that boots a complete functional system within 5 minutes.	*Reduces developer onboarding friction.*

10.2 Configuration Management

Req ID	Priority	Requirement Statement	Rationale

REQ-DEP-010	SHALL	All configuration changes to a running system SHALL be tracked in the audit log: who changed what, from what value to what value, and when.	*Audit trail for operational changes.*
REQ-DEP-011	SHALL	Sensitive configuration values (database passwords, API keys, encryption keys) SHALL be read exclusively from environment variables or mounted secret files; inline config file secrets are prohibited.	*Prevents secrets from appearing in config files or version control.*
REQ-DEP-012	SHALL	The system SHALL provide a configuration validation command that checks all configuration values for correctness without starting the system.	*Enables pre-flight validation in CI/CD pipelines.*
REQ-DEP-013	SHOULD	The system SHOULD support integration with secret management systems (HashiCorp Vault, AWS Secrets Manager) via a pluggable secret backend interface.	*Enterprise-grade secret management integration.*

10.3 Build & Release

Req ID	Priority	Requirement Statement	Rationale
REQ-DEP-020	SHALL	The full Rust build (cargo build --release) SHALL complete in under 20 minutes on a standard 4-core CI runner.	*CI pipeline performance.*
REQ-DEP-021	SHALL	The Java client build (bazel build) SHALL complete in under 10 minutes on a standard 4-core CI runner.	*CI pipeline performance.*
REQ-DEP-022	SHALL	Every release SHALL include: a CHANGELOG entry, updated SBOM (CycloneDX or SPDX), Docker image digest, and GPG-signed release artifacts.	*Release traceability and supply chain integrity.*
REQ-DEP-023	SHALL	Semantic versioning (MAJOR.MINOR.PATCH) SHALL be used for all UME versioning; pre-release versions SHALL use semver pre-release identifiers.	*Predictable version semantics for integrators.*
REQ-DEP-024	SHOULD	The CI/CD pipeline SHOULD include automated security scanning (SAST, dependency vulnerability scan, container image scan) before any release artifact is published.	*Security validation as part of release gate.*

11. Extensibility Requirements

The UME extension model allows third parties and organizations to build custom modules that run inside the UME kernel. These requirements govern the custom module framework.

11.1 Custom Module Framework

Req ID	Priority	Requirement Statement	Rationale
REQ-EXT-001	SHALL	The kernel SHALL provide a stable, versioned module SDK that enables third-party Rust crates to implement the KernelModule trait and register with the kernel.	*Open extensibility is a core UME platform principle.*
REQ-EXT-002	SHALL	Custom modules SHALL be loaded from a configurable module directory or registry; the kernel SHALL hot-load new modules without restart when added to the configured location.	*Zero-downtime module deployment.*
REQ-EXT-003	SHALL	Custom modules SHALL be subject to the same universal contract as built-in modules (REQ-MOD-001 through REQ-MOD-012); no exceptions for custom modules.	*Consistent quality and security standards for all modules.*
REQ-EXT-004	SHALL	Custom UME ecosystem modules SHALL use the namespace custom.ume.{vendor}.{module_name}; organization-private modules SHALL use org.custom.{vendor}.{module_name}.	*Namespace separation prevents ID collisions.*
REQ-EXT-005	SHALL	The system SHALL provide module signature verification: modules MAY be signed; signed modules SHALL have their signature verified before loading; unsigned modules SHALL require explicit opt-in in configuration.	*Supply chain security for the module ecosystem.*
REQ-EXT-006	SHALL	Custom modules SHALL declare their permission namespaces in their ModuleDescriptor; the kernel SHALL register these permissions in the RBAC system automatically on module load.	*Automatic RBAC integration for new modules.*
REQ-EXT-007	SHOULD	The SDK SHOULD provide a module certification tool that validates a module against all universal contract requirements before submission to the UME marketplace.	*Quality gate before marketplace distribution.*
REQ-EXT-008	SHOULD	The UME marketplace SHALL provide discoverability metadata: module name, vendor, version, description, capability tags, compatibility matrix, and installation count.	*Third-party module ecosystem discoverability.*
REQ-EXT-009	MAY	The kernel MAY support WebAssembly (WASM) modules compiled from non-Rust languages as an alternative module runtime with equivalent sandboxing.	*Broader language ecosystem for module development.*

11.2 Integration & Webhook Framework

Req ID	Priority	Requirement Statement	Rationale
REQ-EXT-010	SHALL	The system SHALL support outbound webhooks: any event bus topic may be configured to trigger an HTTP POST to a registered external URL.	*Event-driven integration with any external system.*
REQ-EXT-011	SHALL	Webhook deliveries SHALL include: event payload, event_id, topic, timestamp, and an HMAC-SHA256	*Webhook security and tamper detection.*

		signature using a shared secret; receivers SHALL be able to verify delivery authenticity.	
REQ-EXT-012	SHALL	Failed webhook deliveries SHALL be retried with exponential backoff (max 5 retries); permanently failed deliveries SHALL be logged and available for manual replay.	*Reliable webhook delivery.*
REQ-EXT-013	SHOULD	The system SHOULD support inbound webhook receivers: external systems may POST events to UME endpoints that are translated to internal event bus topics.	*Bidirectional integration without custom module development.*
REQ-EXT-014	SHOULD	The system SHOULD provide a no-code integration builder in the portal: connect event topics to external actions without writing code.	*Enables integration setup by non-developers.*
REQ-EXT-015	MAY	The system MAY provide native integration connectors for common enterprise systems: Slack, Microsoft Teams, Salesforce, SAP, NetSuite, QuickBooks, Xero.	*Reduced integration time for common platforms.*

Appendix A: Requirements Traceability Matrix

The following matrix maps each requirement area to its source stakeholder need and the primary UME components it constrains.

Section	Req Range	Count	Stakeholder Needs Addressed	Primary Components
4. Kernel	REQ-KRN-001–097	40	All — kernel is the universal foundation	UmeKernel, Supervisor, DeviceBus, RBAC, EventBus, Executors, Memory, Storage, Audit
5. Modules	REQ-MOD-001–012 + module-specific	85	All org types by module domain	All 42 organization modules
6. API / Interfaces	REQ-API-001–045	28	Developers, IT Admins, End Users	ume_server, ume_runtime, Java client, Module SDK
7. Data Architecture	REQ-DAT-001–042	19	All — data is shared across all modules	StorageManager, ume_core schemas, MDM Module
8. Security	REQ-SEC-001–044	25	All — particularly regulated industries	SecurityDriver, RBAC Engine, Audit Log, Security Module
9. NFRs	REQ-NFR-001–065	34	All — particularly enterprise and government users	All layers — system-wide quality attributes
10. Deployment	REQ-DEP-001–024	17	IT Admins, System Operators, DevOps	Packaging, CI/CD, Infrastructure, Configuration
11. Extensibility	REQ-EXT-001–015	15	Developers, Organizations with custom needs	Module SDK, Webhook Framework, Marketplace

Total requirements defined in this document: 263

Priority	Count	% of Total
SHALL (Mandatory)	198	75.3%
SHOULD (Recommended)	52	19.8%
MAY (Optional)	13	4.9%

Appendix B: Compliance Standards Reference

Requirements in this document are designed to enable compliance with the following standards. Compliance with a standard is achieved through the combination of UME platform requirements and organizational operational practices.

Standard	Domain	Key UME Requirements
ISO 27001	Information Security Management	REQ-SEC-001 to 044, REQ-KRN-090–097, REQ-NFR-040–043
SOC 2 Type II	Security, Availability, Confidentiality	REQ-SEC-020–025, REQ-NFR-030–037, REQ-KRN-092–096
GDPR / UK GDPR	Data Privacy	REQ-SEC-023, REQ-MOD-SEC-004, REQ-DAT-006, REQ-SEC-043
CCPA	Consumer Privacy (California)	REQ-SEC-023, REQ-MOD-SEC-004, REQ-DAT-006
IFRS / GAAP	Financial Reporting	REQ-MOD-FIN-001–004, REQ-DAT-023
Basel III / IV	Banking Capital Adequacy	REQ-MOD-GRC-001–004, REQ-MOD-RSK-001–006
HIPAA	Healthcare Data	REQ-SEC-020–024, REQ-KRN-092–096, REQ-DAT-010–011
ISO 9001	Quality Management	REQ-MOD-GRC-001–004, REQ-NFR-060–065
ITIL 4	IT Service Management	REQ-MOD-IT-001–004
WCAG 2.1 AA	Web Accessibility	REQ-NFR-053
RFC 2119	Requirements Language	Used throughout for SHALL/SHOULD/MAY definitions

Appendix C: Glossary

Term	Definition
ACID	Atomicity, Consistency, Isolation, Durability — transaction properties required for data integrity.
Audit Record	An immutable, structured record of a system operation used for non-repudiation and compliance.
CMDB	Configuration Management Database — inventory of IT assets and their relationships.
Chombo	UME Legal Entity Management Module (MOD-13); also the compliance policy pack evaluation engine within it.
DeviceBus	The kernel subsystem that registers and manages all device drivers.
DLQ	Dead-Letter Queue — destination for failed event deliveries or process instances.
Domain Event	A fact that has occurred within the system; published to the event bus using {domain}.{resource}.{verb} naming.
Executor Pool	Named thread pools within the kernel for isolating workloads by domain and priority.
KernelModule	The Rust trait that all organization modules must implement to participate in the UME kernel.
KRI	Key Risk Indicator — a measurable metric linked to a risk that signals changing risk exposure.

ModuleDescriptor	A module's self-description: ID, domain, version, dependencies, permissions, and capabilities.
OKR	Objectives and Key Results — goal-setting framework used in the Strategy Module.
RBAC	Role-Based Access Control — the UME authorization model assigning permissions to roles and roles to subjects.
SHALL	RFC 2119 mandatory requirement; non-compliance is a defect.
SHOULD	RFC 2119 strong recommendation; deviation requires documented justification.
MAY	RFC 2119 optional capability; permitted but not required.
Soko	UME Marketing System Module (MOD-22); Swahili for "Market".
Supervisor Engine	The kernel subsystem that monitors module health and applies restart policies.
UME	The Organization / Business Operating System platform described in this document.
ume_core	The shared Rust crate containing all cross-module domain types, traits, and interfaces.